

WILL MCGOWAN

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EDUCATION

University of Strathclyde, Glasgow

Sep 2023 – Jun 2027 (expected)

BA (Hons) Economics

- Average grade to date: 2:1, including 90% in first-year Finance
- Relevant Modules:** Applied Econometrics, Treasury Management and Derivatives, Advanced Microeconomics

Dissertation | MATLAB

In progress, due 2027

"Who Hedges the Barrel? Sovereign Oil Hedging and Petro-Currency Reactions to OPEC News"

- Setting predictions before running any tests: sovereign hedges should make a petro-currency's oil beta smaller (Norway vs unhedged Canada), kinked (Russia's budget-rule cut-off) or one-sided (Mexico's puts)
- Isolating the hedge's effect: each currency's move on OPEC announcement days since 1983, net of the dollar's own move, compared across hedging regimes and against unhedged Canada
- Testing a pre-registered trading rule on any residual beta after costs, with a White reality check and Kelly sizing

Belfast Royal Academy, Belfast

Jun 2023

- A-levels: Mathematics (A), Economics (A), Computer Science (B)

PROJECTS

Plains Heat and the KC-Chicago Wheat Spread

R

- Designed the spread test so shared wheat news cancels, matched weather to prices with no look-ahead and used the off-season as a placebo: Plains heat explained 0.3% of weekly KC-Chicago moves
- Traced a too-good lead-lag signal (0.74 correlation with Chicago's previous day, 0.25 same day) to a one-day timestamp error; with it fixed, a rule set in advance had no edge out of sample

Delta-Hedging Simulator

MATLAB

- Checked the simulator against theory before using it (hedging error should fall as $1/\sqrt{N}$: fitted slope -0.48 vs -0.5), then found 0.1–0.5% trading costs push the best frequency from daily to weekly
- Reasoned that a timetable trades even when the hedge has barely drifted, so tested a no-trade band around delta, chosen on one set of paths and judged on fresh ones: 6% less RMS P&L at 0.1% costs, 2% at 0.5%

Monte Carlo Option Pricing Engine

MATLAB

- Validated the pricer against two closed-form prices first, then picked a geometric-Asian control variate because its payoff tracks the arithmetic one ($\rho = 0.9996$): variance fell 1,276-fold
- Mapped where the control variate weakens over 8 volatilities and 6 strikes and why: its gain falls roughly as $1/\sigma^2$, from 4,569x to 71x at the money as volatility rises from 10% to 80%

Empirical CAPM Test

MATLAB

- Tested the CAPM on 694 anomaly portfolios in 54 countries (JKP data): 215 alphas were significant at 5%, but about 35 would pass by luck alone, so a raw count overstates the evidence
- Applied stricter rules (Benjamini-Hochberg, $|t| > 3$, Bonferroni): 123, 84 and 22 survive, nearly all positive where flukes would split by sign; momentum holds up best, size looks like luck

WORK EXPERIENCE

Le Set, Glasgow | Head Mixologist

Sep 2026 – Present

- Designed a new 200-cover venue's 11-drink menu, including 4 original cocktails; trained around 15 bar staff

KONG, Glasgow | Mixologist and Front of House

Oct 2024 – Jun 2026

- Analysed sales patterns to refine upselling prompts, raising average customer spend by an estimated 25%

LEADERSHIP AND ACTIVITIES

Strathclyde Agricapital Society | Co-Founder and Treasurer

Sep 2025 – Present

- Co-founded Strathclyde's first agri-finance society, grew it to 126 members and published weekly research notes

Strathclyde Poker Society | Vice-President, then President

Sep 2025 – Apr 2026

- Ran weekly tournaments for about 60 players on a £600 budget per semester, reconciling buy-ins and prizes
- Having played 12 tables at once myself, taught new members to use the Kelly criterion to estimate how many buy-ins their win rate, variance and risk level call for, with a buffer in case the edge is overestimated

ADDITIONAL INFORMATION

Technical: R, MATLAB, Excel, Git

Chess: 14th at the 2023 Irish U19 Championships; gave a 22-board simul (average 1860 FIDE) and won 21 games

Powerlifting: Northern Ireland U19 champion (2023)